



Modelling Financial Markets

Seminar 8: Backtesting and Evaluating Risk Forecasts — Hands-on



Daniel Traian PELE

Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ **ACF**: AutoCorrelation Function
- ▣ **ACI**: Adaptive Conformal Inference
- ▣ **AI**: Artificial Intelligence
- ▣ **BCBS**: Basel Committee on Banking Supervision
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BNR**: Banca Națională a României — *the National Bank of Romania*
- ▣ **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- ▣ **CLT**: Central Limit Theorem
- ▣ **DGP**: Data-Generating Process
- ▣ **DM**: Diebold–Mariano (test)
- ▣ **DQ**: Dynamic Quantile (test)
- ▣ **EGARCH**: Exponential GARCH
- ▣ **ES**: Expected Shortfall
- ▣ **ESR**: Expected Shortfall Regression (backtest)
- ▣ **FHS**: Filtered Historical Simulation
- ▣ **FRTB**: Fundamental Review of the Trading Book
- ▣ **FZ0**: Fissler–Ziegel loss, zero-homogeneous version

Acronyms used in this material (2/2)

- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GARCH-EVT**: GARCH filter combined with Extreme Value Theory for the tail
- ▣ **GJR**: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ **GPD**: Generalised Pareto Distribution
- ▣ **GW**: Giacomini–White (test)
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HS**: Historical Simulation
- ▣ **LR**: Likelihood Ratio (test)
- ▣ **LRV**: Long-Run Variance
- ▣ **MCS**: Model Confidence Set
- ▣ **MLE**: Maximum Likelihood Estimation
- ▣ **PIT**: Probability Integral Transform
- ▣ **POF**: Proportion Of Failures (test)
- ▣ **US**: United States
- ▣ **UTC**: Coordinated Universal Time (Temps Universel Coordonné)

Today's Questions and Route

- ▣ Questions of the seminar
 - ▶ **which forecasting procedure would you provisionally choose for VaR 1% and ES 2.5%?**
 - ▶ **how strong is the evidence?**
- ▣ The seminar comes before Lecture 8: start with the section “What You Need for Today” (six slides); it gives every test and score the in-class tasks use
- ▣ Route in class
 - ▶ Step 1: A1, A3 (counting and timing breaches on paper), A5, A7 (traffic lights for VaR and ES), B1 (six VaR models on the S&P 500)
 - ▶ Step 2: B5 (comparing forecasts with HAC standard errors), C2 (critique of an AI answer)
- ▣ After the lecture, at home: B2 (traffic light on the BET), B3 (ES backtests by bootstrap and simulation), B7 (conformal VaR in crises)
- ▣ What you produce: the A answers, one table and one figure per B exercise, the corrected C2 code, one interpretation sentence per exercise
- ▣ Notebook for today: [open the seminar notebook in Colab](#); each task names its notebook section
- ▣ Seminar: Prof. Daniel Traian Pele, danpele@ase.ro; Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Exercise Map

Exercise	Question	Type	Worked example
A1	is 7 breaches in 250 days evidence against a VaR 1%?	Solved	–
A3	do the BET breaches of 2007–2010 cluster?	Solved	–
A5, A7	traffic lights for VaR counts and for ES (Z_2)	Solved	–
B1	which of six VaR models passes on the S&P 500?	Solved	–
B5	is FHS significantly better?	Solved	–
C2	what is wrong in an AI answer?	Proposed	A3, A5
B2	would the supervisor have penalised the GARCH-t model on the BET? (after the lecture)	Proposed	A5, B1
B3	is ES underestimated? (after the lecture)	Solved	–
B7	does conformal VaR keep its coverage in crises? (after the lecture)	Solved	–
Extensions	A1 Extended, A2, A4, A6, A8–A10, B4, B6, B8, B9, C1, C3 (optional, after the seminar)	Solved and Proposed	named on each task

- ▣ **[Solved]**: full solution in the slides and in the notebook, a worked example to follow; **[Proposed]**: you solve it, the solution is discussed in class
- ▣ Part A: derivations on paper; Part B: verification and inference on real forecasts, each ending with an interpretation question; Part C: open questions and AI critique

Data Used and the Forecast Panel

Market	Series	Calendar	Evaluation days
S&P 500	S&P 500 index, daily close	US trading days	03.01.2007–18.09.2026, 4 959
BET	BET index, daily close	BVB trading days	03.01.2007–18.09.2026, 4 937
Bitcoin	Bitcoin in USD, daily close	every day, 24:00 UTC	14.06.2017–18.09.2026, 3 384
EUR/RON	BNR reference rate	Romanian working days	11.06.2009–18.09.2026, 4 352

- Daily closing prices of the course data, one series per market, loaded automatically by the notebook; losses $L_t = -\ln(P_t/P_{t-1})$ in **decimals** ($0.01 = 1\%$), each series on its own calendar, no filling of missing days
- Forecast panel: for each market and model, one row per evaluation day with the loss L_t , VaR 1%, 2.5% and 5%, ES 2.5%, the forecast volatility σ_t , the mean μ_t and the PIT u_t ; the six models of [Chapter 8](#), 1000-day rolling window, refit every 20 days, data up to $t - 1$ only
- The panel is supplied: in class you test forecasts, you do not re-estimate models (optional: the estimation code in the notebook)

Setup: Run and Verify

```
F, tails = fc('sp500')
g = F['GARCH-t']
print(len(g), g.index[0], g.index[-1])
# loss, VaR 1%, VaR 2.5%, ES 2.5%, scale
print(g.iloc[:3, [-1, 0, 1, 3, 6]])
```

- The notebook sections “Data and forecast panel” and “Common helpers” define `fc`, `hits`, `kupiec`, `christoffersen`, `duration_test`, `dq_test`, `fz0`
- Expected output: 4959 rows, 03.01.2007 to 18.09.2026
- A negative L is a gain; VaR and ES are positive loss thresholds
- If your count or first row differs, stop and check the market, the start date and the model name

date	L	VaR1	VaR2.5	ES2.5	scale
2007-01-03	0.001199	0.012031	0.009944	0.012132	0.005298
2007-01-04	-0.001228	0.011894	0.009830	0.011995	0.005241
2007-01-05	0.006103	0.011745	0.009706	0.011844	0.005178

What You Need for Today (1/6): Breaches

- This seminar comes before the lecture: these six slides give what A1, A3, A5, A7, B1, B5 and C2 need
 - ▶ depth in [Chapter 8](#) (the lecture after the seminar), sections “Backtesting VaR: Coverage and Independence” and “The Regulator’s View: Basel Traffic Light and FRTB”
 - ▶ and sections “Backtesting Expected Shortfall” and “Comparative Backtesting”
- A backtest compares out-of-sample forecasts, issued at $t - 1$, with the realised losses
 - ▶ Breach (hit): $I_t = 1\{L_t > \text{VaR}_t\}$; $1\{\cdot\}$: indicator, 1 if the loss L_t exceeds the forecast VaR_t , 0 otherwise
 - ▶ α : the tail probability of the VaR (1%, 2.5%); a correct VaR is breached on a share α of days
- For a correct VaR at level α the hits are i.i.d. Bernoulli(α): independent 0/1 variables equal to 1 with probability α
 - ▶ Two consequences: the right *number* of breaches, and no *memory* (a breach today says nothing about tomorrow)
- The panel holds six procedures ([Chapter 7](#)): HS, Normal and Student-t on a window; GARCH-t, FHS and GARCH-EVT, which filter volatility first

What You Need for Today (2/6): the Kupiec Test

- Kupiec unconditional coverage [Kupiec \(1995\)](#): is the observed breach rate compatible with α ?
 - ▶ x breaches in T days, observed rate $\hat{\pi} = x/T$; $H_0 : \pi = \alpha$, with π the true breach probability
 - ▶ $\ell(\pi) = (T - x) \ln(1 - \pi) + x \ln \pi$: the Binomial log-likelihood of the count if the breach probability were π
- Likelihood ratio: how much better the observed rate explains the count than α does
 - ▶ $LR_{uc} = 2[\ell(\hat{\pi}) - \ell(\alpha)] \approx \chi_1^2$ under H_0
 - ▶ $LR_{uc} \geq 0$; χ_1^2 : chi-square with 1 degree of freedom (one restriction, $\pi = \alpha$); reject at 5% if $LR_{uc} > 3.84$
 - ▶ The LR principle is the one of [Chapter 5](#) (GJR against GARCH); here the model is a Binomial count
- Example: $T = 250$, $\alpha = 1\%$: 2.5 breaches expected
 - ▶ $x = 2$ gives $LR_{uc} = 0.11$: no evidence against the model
 - ▶ with so few expected breaches, the exact Binomial p is safer than χ_1^2

What You Need for Today (3/6): Independence

- Christoffersen [Christoffersen \(1998\)](#): a two-state Markov chain for the hits (state 0 = no breach, 1 = breach)
 - ▶ $\pi_{01} = P(I_t = 1 \mid I_{t-1} = 0)$, $\pi_{11} = P(I_t = 1 \mid I_{t-1} = 1)$: the breach probability after a calm day and after a breach
 - ▶ n_{ij} : number of days in state j after a day in state i ; estimates $\hat{\pi}_{01} = n_{01}/(n_{00} + n_{01})$, $\hat{\pi}_{11} = n_{11}/(n_{10} + n_{11})$
 - ▶ Example: 12 days 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0: pairs (I_{t-1}, I_t) give $n_{00} = 6$, $n_{01} = 2$, $n_{10} = 2$, $n_{11} = 1$
- Two likelihood ratio tests
 - ▶ LR_{ind} : $H_0 : \pi_{01} = \pi_{11}$ (no memory), one restriction, χ_1^2 ; large when a breach makes the next breach more likely
 - ▶ $\text{LR}_{cc} = \text{LR}_{uc} + \text{LR}_{ind}$: $H_0 : \pi_{01} = \pi_{11} = \alpha$ (right rate and no memory), two restrictions, χ_2^2 (5% critical value 5.99)

What You Need for Today (4/6): Durations and DQ

- Duration test [Christoffersen & Pelletier \(2004\)](#): the spacing of breaches
 - ▶ Under H_0 the numbers of days between breaches are geometric (memoryless), with mean $1/\alpha$
 - ▶ Fitted model: a Weibull distribution with shape b ; $b = 1$ is memoryless, $b < 1$ means clusters of breaches followed by long silences
- Dynamic quantile (DQ) test [Engle & Manganelli \(2004\)](#): can anything known yesterday predict today's hit?
 - ▶ Regress the centred hit $I_t - \alpha$ on a constant, lagged hits and VaR_t ; under H_0 all coefficients are zero
 - ▶ Wald test of that hypothesis, χ^2 with as many degrees of freedom as regressors; large DQ = predictable hits
- With one year of data these tests have a size away from 5% and little power: B1 reads them with simulated sizes

What You Need for Today (5/6): The Traffic Light and an ES Backtest

□ Basel traffic light BCBS (1996)

- ▶ count the exceptions X of VaR 1% in the last 250 days
- ▶ correct model: $X \sim \text{Binomial}(250, 0.01)$, breaches in 250 independent days with probability 1% each
- ▶ green while $P(X \leq x) < 95\%$, red from $P(X \leq x) \geq 99.99\%$, yellow in between
- ▶ yellow and red raise the capital multiplier (1996: 3 plus a factor; FRTB BCBS (2019): from 1.5 to 2)
- ▶ A zone is a lookup rule for a supervisor, not a 5% test

□ ES backtest Z_2 Acerbi & Szekely (2014): total breach loss in units of the forecast ES, against T_α

- ▶ $Z_2 = 1 - \frac{1}{T_\alpha} \sum_t \frac{L_t I_t}{\text{ES}_t}$, with the breaches of VaR at the same level α (2.5%); ES_t : the ES forecast for day t
- ▶ correct ES: breach losses average ES_t and about T_α breaches occur, so $E[Z_2] = 0$ (proof in A7)
- ▶ $Z_2 < 0$: too many or too large tail losses
- ▶ Example: $T_\alpha = 6.25$, seven breaches, each with $L_t/\text{ES}_t = 1.1$: $Z_2 = 1 - 7.7/6.25 = -0.23$

What You Need for Today (6/6): Scores and the Diebold–Mariano Test

- **Consistent score** [Gneiting \(2011\)](#): expected value smallest at the true VaR (and ES)
 - ▶ a tail forecast cannot be compared by a squared error
 - ▶ VaR: pinball loss $S(v, L) = (1\{L > v\} - \alpha)(L - v)$, v : the VaR forecast
 - ▶ e.g. $\alpha = 1\%$, $v = 3$: a loss $L = 5$ scores $0.99 \times 2 = 1.98$, a gain $L = -1$ scores $0.01 \times 4 = 0.04$
 - ▶ (VaR, ES): FZ0 [Patton, Ziegel & Chen \(2019\)](#) = $\frac{1\{L > v\}(L - v)}{\alpha e} + \frac{v}{e} + \ln e - 1$, with $v = \text{VaR}$, $e = \text{ES} > 0$; lower is better
- Diebold–Mariano [Diebold & Mariano \(1995\)](#) (Chapter 5): is the average score difference of models A and B significantly different from 0?
 - ▶ $d_t = S_{A,t} - S_{B,t}$ (score of A minus score of B on day t), $\bar{d}$: its mean; $\text{DM} = \bar{d} / \sqrt{\widehat{\text{LRV}}/T} \approx N(0, 1)$ under equal expected scores; $\text{DM} > 1.96$: B significantly better
 - ▶ d_t is autocorrelated, so the long-run variance replaces γ_0 : Newey–West [Newey & West \(1987\)](#)
 $\widehat{\text{LRV}} = \gamma_0 + 2 \sum_{k=1}^q (1 - \frac{k}{q+1}) \gamma_k$, γ_k the autocovariances of d_t at lag k , q the number of lags
 - ▶ Example, $q = 1$: $\gamma_0 = 4$, $\gamma_1 = 1$: $\widehat{\text{LRV}} = 4 + 2 \times 0.5 \times 1 = 5$; with $\bar{d} = 0.1$, $T = 2000$: $\text{DM} = 0.1 / \sqrt{5/2000} = 2.0$

Notation, and What Is Known When a Forecast Is Issued

- ▣ Loss $L_t = -r_t$; forecasts VaR_t , ES_t , σ_t are issued at the close of day $t - 1$ and use only $\mathcal{F}_{t-1}$ (returns up to r_{t-1})
- ▣ Level = tail probability α : VaR 1% ($\alpha = 0.01$), VaR 2.5% and ES 2.5% ($\alpha = 0.025$); breach $I_t = 1\{L_t > \text{VaR}_t\}$, target rate α
- ▣ T : evaluation days; $x = \sum_t I_t$: breaches; $\hat{\pi} = x/T$; n_{ij} : days in state j after a day in state i
- ▣ H_0 of every backtest: $P(I_t = 1 \mid \mathcal{F}_{t-1}) = \alpha$, i.e. the I_t are i.i.d. Bernoulli(α); a test checks one implication (count, memory, spacing)
- ▣ Scores for comparing forecasts: pinball for VaR, FZ0 for (VaR, ES); lower is better
- ▣ One worked line: $T = 250$, $x = 7$: $\hat{\pi} = 2.8\%$ against $\alpha = 1\%$, 2.5 breaches expected

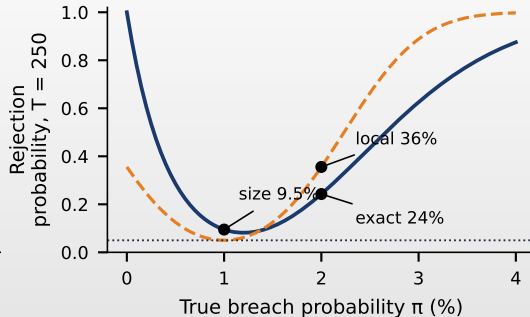
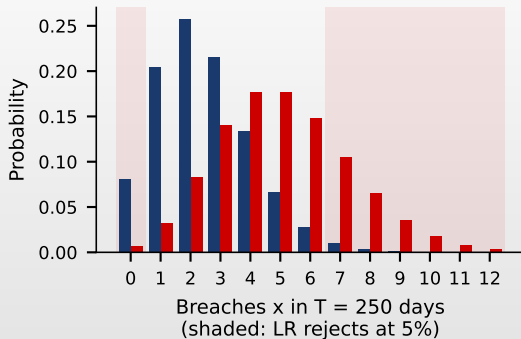
A1 [Solved]: Kupiec from a Breach Count — Task

- ▣ **Question:** is $x = 7$ breaches in $T = 250$ days evidence against a VaR 1% model?
- ▣ **Inputs:** $T = 250$, $x = 7$, $\alpha = 0.01$; the hits are i.i.d. Bernoulli(π); no data file
- ▣ **Tasks**
 1. Compute $\hat{\pi} = x/T$ and the log-likelihoods $\ell(\alpha) = (T - x) \ln(1 - \alpha) + x \ln \alpha$ (restricted) and $\ell(\hat{\pi})$ (unrestricted).
 2. Compute $\text{LR}_{uc} = 2[\ell(\hat{\pi}) - \ell(\alpha)]$, its χ^2_1 p -value, and the exact one-sided probability $P(X \geq 7)$ under Binomial(250, 0.01).
 3. List the integers x with $\text{LR}_{uc}(x) \leq 3.84$: the acceptance region of the rule “reject if $\text{LR}_{uc} > 3.84$ ”.
 4. Sum Binomial probabilities outside that region to get the actual size ($\pi = 1\%$) and the power against $\pi = 2\%$; compare with the local power $P(\chi^2_1(\lambda) > 3.84)$, $\lambda = T(\pi - \alpha)^2/(\alpha(1 - \alpha))$.
- ▣ **Report:** $\hat{\pi}$, the two log-likelihoods, LR_{uc} with both p -values, the acceptance region, nominal size, actual size and the two powers
- ▣ **Interpretation:** can one year of VaR 1% data detect a model that doubles the breach rate?
- ▣ Notebook section: A1

A1 [Solved]: Solution (1/3) — The Test Step by Step

1. $\hat{\pi} = 7/250 = 0.028$; $\ell(\alpha) = 243 \ln 0.99 + 7 \ln 0.01 = -34.678$; $\ell(\hat{\pi}) = 243 \ln 0.972 + 7 \ln 0.028 = -31.930$.
 2. $LR_{uc} = 2(-31.930 - (-34.678)) = 5.50$; χ_1^2 $p = 0.019$; exact $P(X \geq 7) = 0.014$: both reject at 5% (too many breaches).
 3. Evaluating $LR_{uc}(x)$ for $x = 0, 1, 2, \dots$: $LR_{uc}(0) = -500 \ln 0.99 = 5.03 > 3.84$, $LR_{uc}(6) = 3.56$, $LR_{uc}(7) = 5.50$: the acceptance region is $\{1, \dots, 6\}$.
 4. Three different quantities: **nominal size** 5% (the χ_1^2 critical value), **actual size** of this rule (next slide), **power** against $\pi = 2\%$ (next slide).
- ☐ Interpretation: 7 breaches are 2.8 times the 2.5 expected; a correct VaR 1% produces 7 or more breaches in only 0.014 of years

A1 [Solved]: Solution (2/3) — Size and Power



■ $P(x)$, correct model, $\pi = 1\%$ ■ $P(x)$, true $\pi = 2\%$ — Exact rejection probability - - Local asymptotic power

A1 [Solved]: Solution (3/3) — Size and Power

- ▣ Actual size: $P(X = 0) + P(X \geq 7) = 8.11\% + 1.37\% = 9.5\%$ under $\pi = 1\%$, not 5%
- ▣ Exact power at $\pi = 2\%$: $1 - P(1 \leq X \leq 6) = 1 - 75.7\% = 24\%$
- ▣ Local: $\lambda = 250 \cdot 0.01^2 / 0.0099 = 2.53$, $P(\chi_1^2(\lambda) > 3.84) = 36\%$: the asymptotics overstate the power
- ▣ Interpretation: with 2.5 expected breaches the lattice of x dominates; a model that doubles the breach rate passes one year of testing about three times in four

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A3 [Solved]: From a Hit Sequence to Transitions — Task

▣ Questions

- ▶ do the breaches of historical simulation on the BET in 2007–2010 cluster?
- ▶ are there simply too many of them?

▣ Inputs

- ▶ Teaching sequence of 12 days: $I_t = 0\ 0\ 1\ 1\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0$
- ▶ BET daily closes, HS VaR 1% (1000-day window), 2007–2010: $T = 1005$ days, so $n = T - 1 = 1004$ transitions (the first day is conditioned on)
- ▶ Transition counts: $n_{00} = 958$, $n_{01} = 22$, $n_{10} = 22$, $n_{11} = 2$

▣ Tasks

1. Count the 11 transitions of the 12-day sequence into a 2×2 table $(n_{00}, n_{01}, n_{10}, n_{11})$.
2. For the BET, compute $\hat{\pi}_{01} = n_{01}/(n_{00} + n_{01})$, $\hat{\pi}_{11} = n_{11}/(n_{10} + n_{11})$ and $\hat{\pi} = (n_{01} + n_{11})/n$.
3. Write the restricted log-likelihood $\ln L(\hat{\pi}, \hat{\pi})$ and the unrestricted $\ln L(\hat{\pi}_{01}, \hat{\pi}_{11})$; compute LR_{ind} and its χ^2_1 p -value.
4. Compute LR_{uc} on the same n transitions ($x = 24$), then $\text{LR}_{cc} = \text{LR}_{uc} + \text{LR}_{ind}$ and its χ^2_2 p -value.

▣ **Report:** the toy table, three probabilities, LR_{ind} , LR_{uc} , LR_{cc} with p -values

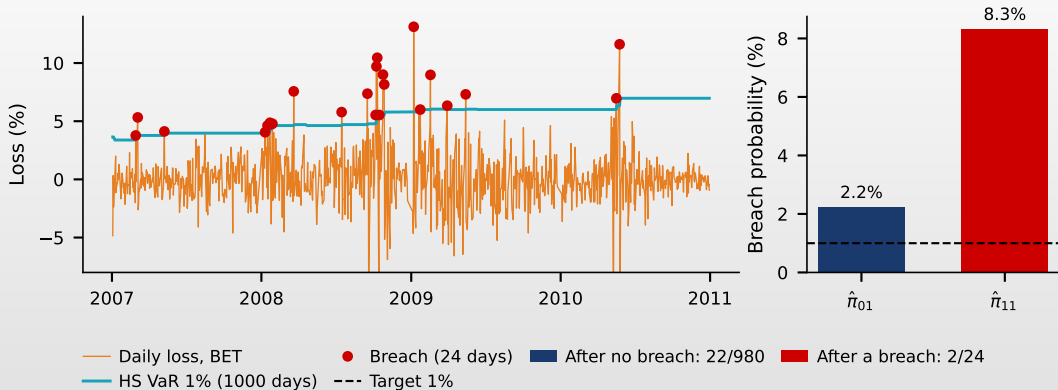
▣ **Interpretation:** does HS fail because of too many breaches or because breaches cluster?

▣ Notebook section: A3

A3 [Solved]: Solution (1/3) — Counts and Likelihoods

1. Toy sequence: pairs (I_{t-1}, I_t) give $n_{00} = 6$, $n_{01} = 2$, $n_{10} = 2$, $n_{11} = 1$ (total 11).
 2. BET: $\hat{\pi}_{01} = 22/(958 + 22) = 0.0224$; $\hat{\pi}_{11} = 2/(22 + 2) = 0.0833$; $\hat{\pi} = (22 + 2)/1004 = 0.0239$.
 3. $\ln L(\pi_{01}, \pi_{11}) = n_{00} \ln(1 - \pi_{01}) + n_{01} \ln \pi_{01} + n_{10} \ln(1 - \pi_{11}) + n_{11} \ln \pi_{11}$; each part is a Bernoulli likelihood, so the MLEs are the ratios of step 2; $\text{LR}_{ind} = 2.32$, $p = 0.128$: independence not rejected.
 4. $\text{LR}_{uc} = 14.11$ on the 1004 transitions; $\text{LR}_{cc} = 14.11 + 2.32 = 16.43$, $p = 0.0003$ (χ^2_2 : both π_{01} and π_{11} fixed at α): rejected.
- On the same transitions the decomposition $\text{LR}_{cc} = \text{LR}_{uc} + \text{LR}_{ind}$ is exact; with LR_{uc} on all T days it holds only asymptotically

A3 [Solved]: Solution (2/3) — What the Chart Shows



A3 [Solved]: Solution (3/3) — What the Chart Shows

- The HS VaR moves in steps and stays flat while losses cluster in late 2008: 24 breaches where 10 are expected
- After a breach the breach probability is almost four times higher (0.0833 versus 0.0224), yet it rests on only 2 consecutive pairs
- Interpretation: HS fails mainly through too many breaches ($\hat{\pi} \approx 2.4\%$); the clustering is visible but four years of data cannot confirm it with a first-order test

 MFM_ch8_sem_partA

A5 [Solved]: The Traffic Light from the Binomial — Task

▣ **Questions:** a bank reports 6 exceptions of its VaR 1% in 250 days

- ▶ which zone is it in?
- ▶ what is the capital consequence?
- ▶ how unusual is it?

▣ **Inputs**

- ▶ Correct model: $X \sim \text{Binomial}(250, 0.01)$; green: $P(X \leq x) < 95\%$; red: $\geq 99.99\%$ (BCBS (1996))
- ▶ Table (right): 1996 plus factor on the multiplier 3; FRTB multiplier (BCBS (2019))

Exceptions	1996 plus factor	FRTB multiplier
0–4	0	1.50
5	0.40	1.70
6	0.50	1.76
7	0.65	1.83
8	0.75	1.88
9	0.85	1.92
≥ 10	1.00	2.00

▣ **Tasks**

1. Compute $P(X \leq x)$ for $x = 4, 5, 9, 10$ and find where the zones start.
2. For $x = 6$: the zone, the 1996 and FRTB multipliers, $P(X \geq 6)$.
3. Compute $P(X \leq 4)$ when the true breach probability is 3%.
4. Compare $P(X \geq 5)$ (a correct model leaves green) with red.

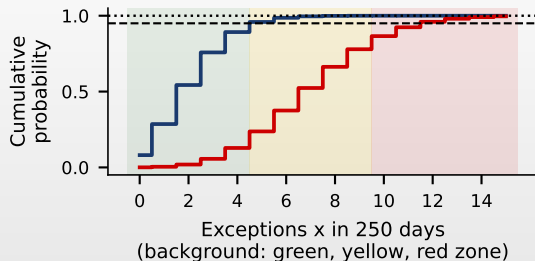
▣ **Report:** four probabilities, the zones, the answers to 2–4

▣ **Interpretation**

- ▶ which error does the traffic light control?
- ▶ which error does it leave large?

▣ Notebook section: A5

A5 [Solved]: Solution



- Zones: green 0–4, yellow 5–9, red 10 or more
 - ▶ $P(X \leq 4) = 89.22\%$, $P(X \leq 5) = 95.88\%$
 - ▶ $P(X \leq 9) = 99.97\%$, $P(X \leq 10) = 99.99\%$
- $x = 6$: yellow
 - ▶ multiplier 1996: $3 + 0.50 = 3.50$; FRTB: 1.76
 - ▶ $P(X \geq 6) = 100\% - 95.88\% = 4.12\%$
- True rate 3%: $P(X \leq 4) = 12.8\%$
 - ▶ a model that triples the breach rate stays green about one year in eight
- A correct model leaves green with $P(X \geq 5) = 10.78\%$
 - ▶ but almost never reaches red: the zones control false red alarms
 - ▶ a zone is a lookup rule, not a 5% significance test

A7 [Solved]: Z_1 and Z_2 on One Year — Task

□ **Question:** do eight breach days in one year show that ES 2.5% is underestimated?

□ **Inputs**

- ▶ $T = 250$ days, $\alpha = 0.025$, so $T\alpha = 6.25$; the table lists the eight days that are **already identified** as breaches of VaR 2.5%
- ▶ $Z_1 = 1 - \frac{1}{N_T} \sum_t \frac{L_t I_t}{ES_t}$, $Z_2 = 1 - \frac{1}{T\alpha} \sum_t \frac{L_t I_t}{ES_t}$ (Acerbi & Szekely (2014))

□ **Tasks**

1. Complete the column L_t/ES_t and compute each day's contribution $L_t/(ES_t T\alpha)$ to the sum in Z_2 .
2. Compute Z_1 and Z_2 ; classify Z_2 with the reference thresholds -0.70 and -1.8 (illustrative; B3 calibrates them for a model).
3. Replace one non-breach day of the same window by a breach with $L = 7\%$, $ES = 3.5\%$ (T stays 250) and recompute Z_2 .
4. Show $E[L_t I_t | \mathcal{F}_{t-1}] = \alpha ES_t$ for a correct continuous forecast and deduce $E[Z_2] = 0$.

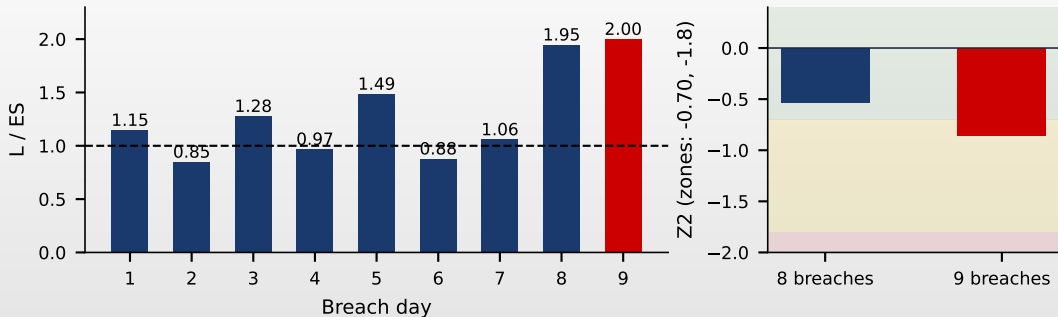
□ **Report:** the ratios, Z_1 , Z_2 before and after, the zone, the proof

□ **Interpretation:** which days drive Z_2 ?

□ Notebook section: A7

Day	L_t (%)	ES_t (%)	L_t/ES_t
1	3.9	3.4	...
2	2.8	3.3	...
3	4.6	3.6	...
4	3.1	3.2	...
5	5.2	3.5	...
6	2.9	3.3	...
7	3.6	3.4	...
8	7.4	3.8	...

A7 [Solved]: Solution (1/2)

--- $L = ES$ ■ L/ES on the 8 breach days■ Added breach: $7/3.5 = 2$ ■ $Z2$, 8 breaches = -0.54■ $Z2$, 9 breaches = -0.86

A7 [Solved]: Solution (2/2)

- ▣ Ratios: 1.147, 0.848, 1.278, 0.969, 1.486, 0.879, 1.059, 1.947; sum 9.613; contributions = ratio/6.25, from 0.136 (day 2) to 0.312 (day 8)
- ▣ $Z_1 = 1 - 9.613/8 = -0.202$: tail losses 20% above ES on average; $Z_2 = 1 - 9.613/6.25 = -0.538 > -0.70$: green
- ▣ Added breach, ratio 2: $Z_2 = -0.86 < -0.70$: yellow
- ▣ Proof: $E[L_t I_t | \mathcal{F}_{t-1}] = P(L_t > \text{VaR}_t | \mathcal{F}_{t-1}) E[L_t | L_t > \text{VaR}_t, \mathcal{F}_{t-1}] = \alpha \text{ES}_t$; $T\alpha$ is constant, so $E[Z_2] = 0$ (Z_1 divides by the random N_T)
- ▣ Interpretation: Z_2 reacts to both the number and the size of breaches; one large loss can change the zone

 MFM_ch8_sem_partA

B1 [Solved]: Six VaR Models on the S&P 500 — Task

Questions

- ▶ which of the six procedures pass the VaR 1% backtests?
- ▶ which test decides?

Inputs

- ▶ Notebook section B1; S&P 500 forecast panel: daily loss L_t and VaR 1% of the six models; $T = 4959$; table: ten GARCH-t days, September 2008
- ▶ Durations: ends censored; DQ regressors: constant, 4 lagged hits, VaR_t

Tasks

1. Fill the column I_t of the table and count the breaches.
2. For GARCH-t in 2008 ($T = 253$): compute x , LR_{uc} , the χ^2_1 and the exact two-sided Binomial p -values; decide at 5%.
3. All six models, full sample: x , rate, Kupiec (asymptotic, exact), independence, CC, Weibull b and duration p -value, DQ p -value.
4. Plot the breach days and the survival curves of the durations.
5. Read a DQ $p = 0.03$ at $T = 250$ against the simulated size of the tests.

- ▣ **Report:** one table row per model (task 3 statistics), two figures

Interpretation

- ▶ which models pass all tests?
- ▶ why can a model with the right count still fail?

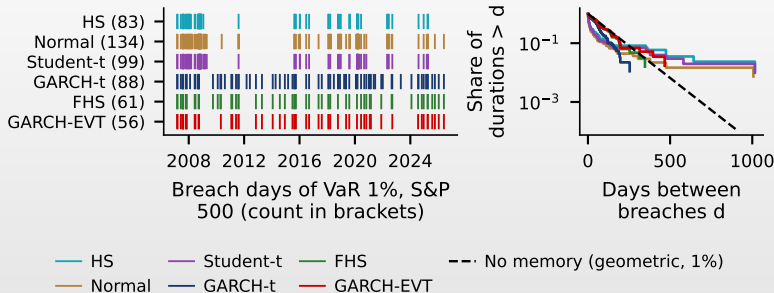
Date	L_t (%)	VaR_t (%)	I_t
04.09.2008	3.04	2.98	?
05.09.2008	-0.44	3.45	?
08.09.2008	-2.03	3.35	?
09.09.2008	3.47	3.46	?
10.09.2008	-0.61	3.99	?
11.09.2008	-1.37	3.88	?
12.09.2008	-0.21	3.84	?
15.09.2008	4.83	3.72	?
16.09.2008	-1.74	4.68	?
17.09.2008	4.83	4.64	?

B1 [Solved]: Solution (1/4) — One Model Step by Step

Date	L_t	VaR_t	I_t
04.09.2008	3.04	2.98	1
05.09.2008	-0.44	3.45	0
08.09.2008	-2.03	3.35	0
09.09.2008	3.47	3.46	1
10.09.2008	-0.61	3.99	0
11.09.2008	-1.37	3.88	0
12.09.2008	-0.21	3.84	0
15.09.2008	4.83	3.72	1
16.09.2008	-1.74	4.68	0
17.09.2008	4.83	4.64	1

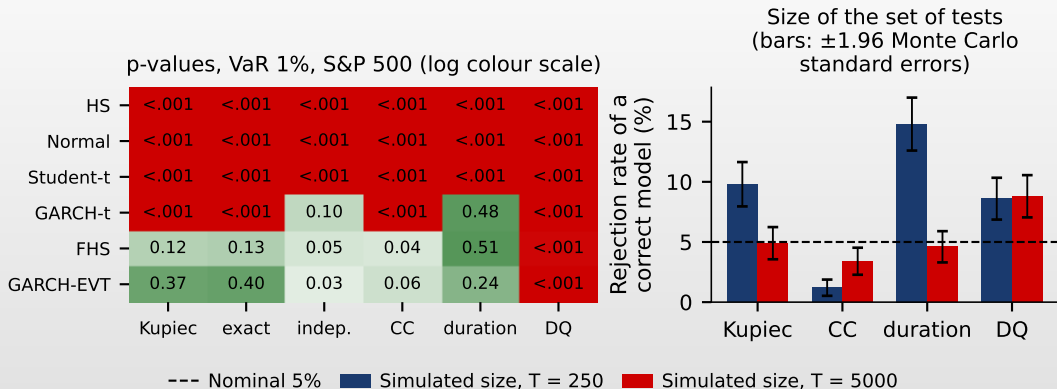
1. $I_t = 1$ when $L_t > \text{VaR}_t$: 4 breaches in 10 days, against 0.1 expected: a burst, while the VaR rises only after each loss.
2. 2008: $x = 8$ in $T = 253$, $\hat{\pi} = 3.16\%$; $\text{LR}_{uc} = 7.60$, $p = 0.0058$; exact $p = 0.0086$: rejected at 5%.
3. Full sample: the same hit series goes into all five tests; the counts give the rates, the transitions give independence, the gaps between breaches give the durations.
4. With 4 959 days the asymptotic and exact Kupiec p -values agree; in one year they differ (0.0058 versus 0.0086).

B1 [Solved]: Solution (2/4) — When the Breaches Happen



- Unconditional models (HS, Normal, Student-t): bursts in 2008–2009, 2020 and 2022, then years of silence; Weibull b : HS 0.52, Normal 0.56, Student-t 0.53 (duration $p < 0.001$)
- Filtered models: survival curves close to the geometric line; GARCH-t 0.94 ($p = 0.479$), GARCH-EVT 0.88 ($p = 0.240$)

B1 [Solved]: Solution (3/4) — All Tests and Their Size



B1 [Solved]: Solution (4/4) — All Tests and Their Size

- Only GARCH-EVT passes Kupiec, CC and durations; FHS fails CC narrowly ($p = 0.040$); DQ rejects all six (GARCH-EVT $p < 0.001$): hits are predictable from lagged hits and the VaR level
- Size at $T = 250$: POF 9.8%, CC 1.2%, durations 14.8%, DQ 8.6% (Monte Carlo standard error about 0.9 points): a DQ $p = 0.03$ in one year is weak evidence; simulate the critical value
- Interpretation: a right count (FHS: 1.23%) can still fail on timing; only filtered models with a flexible tail survive most tests

 MFM_ch8_sem_var

B5 [Solved]: Is FHS Significantly Better? — Task

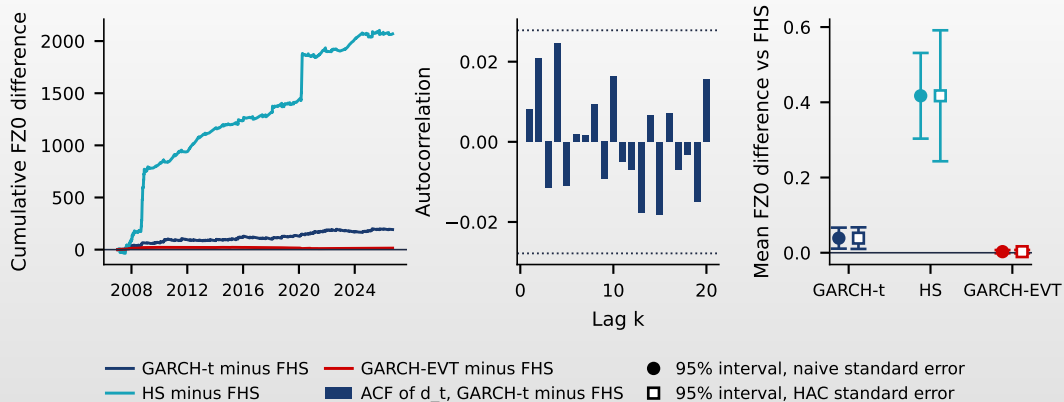
- **Question:** is FHS significantly better than GARCH-t, HS and GARCH-EVT under the FZ0 score on the S&P 500?
- **Inputs**
 - ▶ Daily FZ0 scores of (VaR 2.5%, ES 2.5%) for the six models on the S&P 500, aligned on the same 4 959 days ([Patton, Ziegel & Chen \(2019\)](#))
 - ▶ Score differential $d_t = S_{A,t} - S_{FHS,t}$ (a difference of scores, not of losses); Newey–West with $q = \lfloor 4(T/100)^{2/9} \rfloor = 9$ lags, $\lfloor \cdot \rfloor$: integer part ([Newey & West \(1987\)](#))
- **Tasks**
 1. Compute the mean FZ0 of each model.
 2. For $A = \text{GARCH-t}$: $\bar{d}$, the autocovariances $\gamma_0, \dots, \gamma_9$, $\text{LRV} = \gamma_0 + 2 \sum_{k=1}^9 (1 - k/10) \gamma_k$, $\text{se} = \sqrt{\text{LRV}/T}$, $\text{DM} = \bar{d}/\text{se}$ and its p -value ([Diebold & Mariano \(1995\)](#)).
 3. Repeat for HS and GARCH-EVT; compute the naive t (ordinary standard error) and the ratio LRV/γ_0 .
 4. Plot the cumulative d_t , the ACF of d_t and the naive and HAC 95% intervals of $\bar{d}$.
 5. Optional: the Giacomini–White test with $h_{t-1} = (1, d_{t-1})$ ([Giacomini & White \(2006\)](#)).
- **Report:** mean scores, one complete DM calculation, a DM table and one figure
- **Interpretation**
 - ▶ which differences are significant?
 - ▶ when does ignoring autocorrelation overstate significance?
- Notebook section: B5

B5 [Solved]: Solution (1/3) — One DM Statistic Step by Step

1. Mean FZ0: FHS -3.551 , GARCH-EVT -3.548 , GARCH-t -3.512 , HS -3.134 , Student-t -3.036 , Normal -2.958 .
2. GARCH-t minus FHS: $\bar{d} = 0.0388$; $\gamma_0 = 1.0076$; $\gamma_1, \dots, \gamma_9$: $0.0081, 0.0210, -0.0116, 0.0249, -0.0110, 0.0018, 0.0015, 0.0094, -0.0094$.
3. $2 \sum_k (1 - k/10) \gamma_k = 0.0551$, so $LRV = 1.0627$; $se = \sqrt{1.0627/4959} = 0.0146$ (naive 0.0143).
4. $DM = 0.0388/0.0146 = 2.65$, $p = 0.008$: GARCH-t is significantly worse at 5%.

A minus FHS	$\bar{d}$	DM	p	naive t	LRV/γ_0
GARCH-t	0.0388	2.65	0.008	2.72	1.05
HS	0.4173	4.70	<0.001	7.18	2.34
GARCH-EVT	0.0029	1.24	0.216	1.31	1.11

B5 [Solved]: Solution (2/3) — Differentials, ACF and Intervals



B5 [Solved]: Solution (3/3) — Differentials, ACF and Intervals

- HS loses steadily and in jumps (2008, 2020); its LRV is 2.34 times γ_0 , so the naive t (7.18) overstates the HAC DM (4.70); for GARCH-t the ACF is small (lag 1: 0.008) and the two intervals almost coincide
- Provisional ranking: FHS and GARCH-EVT tie ($p = 0.216$); GARCH-t and HS are significantly worse; GW (optional): GARCH-EVT 7.38 ($p = 0.025$): the conditional tie fails, which moment drives it needs a separate test

 MFM_ch8_sem_compare

C2 [Proposed]: Critique an AI Answer — Task

▣ Questions

- ▶ is the AI answer on the next slide correct?
- ▶ does its conclusion survive once the code is fixed?

▣ Worked example to follow: A3, A5

▣ Inputs

- ▶ The prompt, the answer and its line-numbered code (next slide); `lr_ind(I)` is `christoffersen(I, 0.01) ['LR_ind']` from the notebook helpers
- ▶ Test series: 401 prices from 100, daily log returns $0.01 \times t_4 / \sqrt{2}$ (seed 8) with $r = -8\%$ at position 320; real data: S&P 500 daily closes from 1990

▣ Tasks

1. Find the three conceptual errors: in the forecast window, in the test distribution, in the regulatory reading.
2. For each error, say how you would detect it: by a numerical check of the output, by the formulas of [Chapter 8](#), or by a published table (BCBS).
3. Run the answer's code and your corrected code on the test series and on the S&P 500; report VaR at position 320 and the exception counts.
4. Give the correct zone, plus factor and multiplier for 7 exceptions (A5).
5. Optional checks: the evaluation window of the traffic light; $x = 0$ in the Kupiec formula.

▣ Report: the three errors with detection and correction, the corrected code, two exception counts

▣ Interpretation: does “the model must be withdrawn” survive?

▣ Notebook section: C2

C2 [Proposed]: Critique an AI Answer — the Answer

- **Prompt:** “I have daily S&P 500 closes in a pandas Series P. Write Python for a rolling 250-day historical-simulation VaR 1%, count the exceptions and run the Kupiec and Christoffersen tests. My last 250 days have 7 exceptions: what does that mean?”

```
1 import numpy as np, scipy.stats as stats
2 r = np.log(P).diff().dropna().values
3 W, p = 250, 0.01
4 VaR = np.array([-np.quantile(r[t-W:t+1], p) for t in range(W, len(r))])
5 I = (-r[W:] > VaR).astype(int)
6 T, x = len(I), I.sum(); ph = x / T
7 LR_uc = -2*((T-x)*np.log(1-p) + x*np.log(p) - (T-x)*np.log(1-ph) - x*np.log(ph))
8 LR_cc = LR_uc + lr_ind(I)
9 p_uc, p_cc = stats.chi2.sf(LR_uc, df=1), stats.chi2.sf(LR_cc, df=1)
```

- **Answer:** “Your last 250 days: $x = 7$, $LR_{uc} = 5.50$, $p = 0.019$: coverage rejected at 5%. Under the Basel traffic light 7 exceptions is the red zone, so the model must be withdrawn.”
- AI code usually runs without an error message: the mistakes are in the finance and the statistics, not in the syntax

After the Lecture: What Comes Next

- B3 and B7 use tools that Lecture 8 develops: residual bootstrap and simulation tests for ES (B3), conformal prediction (B7); read them after the lecture
- Depth: [Chapter 8](#), sections “Backtesting Expected Shortfall” and “Conformal Prediction for Tail Risk”
- Both are solved: work through them as worked examples before the homework B2 and the extension packs B4, B8

B3 [Solved]: Is ES Underestimated? — Task (1/2)

- **Question:** do the ES 2.5% forecasts of GARCH-t, FHS and HS understate the tail of the S&P 500?
- **Inputs**
 - ▶ S&P 500 forecast panel: loss L_t , VaR 2.5%, ES 2.5%, forecast volatility σ_t (for HS the window standard deviation), and for HS and FHS the 25 largest window losses of each day; $T = 4959$
 - ▶ One day step by step, GARCH-t, 12.03.2020: $L = 9.99\%$, $\text{VaR} = 9.40\%$, $\text{ES} = 13.43\%$, $\sigma = 4.81\%$
- **Tasks**
 1. For that day compute the exceedance residual $e_t = (L_t - \text{ES}_t)/\sigma_t$ and the contribution $L_t/(\text{ES}_t T\alpha)$ to the sum in Z_2 .
 2. Run Algorithm 1 (next slide) for the three models: n , $\bar{e}$, t , bootstrap p and Student- t p .
 3. Run Algorithm 2 for the three models on the full sample and on the last 250 days: Z_2 , p , simulated 5% critical value.
 4. Plot the two null distributions of Z_2 (GARCH-t) and the bootstrap distribution of t .
- **Report:** one table (model, n , $\bar{e}$, t , two p , Z_2 , p , critical value, both samples) and one figure
- **Interpretation:** what does a rejection say about the joint forecasting procedure?
- Notebook section: B3

B3 [Solved]: Two Algorithms — Task (2/2)

Algorithm 1: McNeil–Frey residual bootstrap

1. Keep the breach days $L_t > \text{VaR}_t$; $e_t = (L_t - \text{ES}_t)/\sigma_t$, n residuals.
2. $t = \bar{e}/(s_e/\sqrt{n})$ ($\bar{e}$, s_e : mean and standard deviation of the residuals), $H_1: E[e_t] > 0$.
3. Centre: $\tilde{e}_t = e_t - \bar{e}$ (imposes H_0).
4. 5000 times: draw n values of $\tilde{e}$ with replacement (seed 0), compute t^* .
5. $p = \text{share of } t^* \geq t$; assumes independent residuals.

Algorithm 2: Z_2 under the model's own forecasts

1. Hold every VaR_t , ES_t fixed; Z_2 from the data.
2. 5000 times (seed 11), each day: $U_t \sim U(0, 1)$; if $U_t > 1 - \alpha$, $L_t^* = F_t^{-1}(U_t)$, else no breach.
3. F_t^{-1} : the forecast quantile function of L_t ; GARCH- $t - \mu_t + \sigma_t t_\nu^{-1}(U) \sqrt{(\nu - 2)/\nu}$ (t_ν^{-1} : Student- t quantile, rescaled to variance 1); HS, FHS: the stored tail losses.
4. $Z_2^* = 1 - \sum_t L_t^* 1\{U_t > 1 - \alpha\} / (\text{ES}_t T \alpha)$.
5. $p = \text{share } Z_2^* \leq Z_2$; critical value: 5% quantile of Z_2^* .

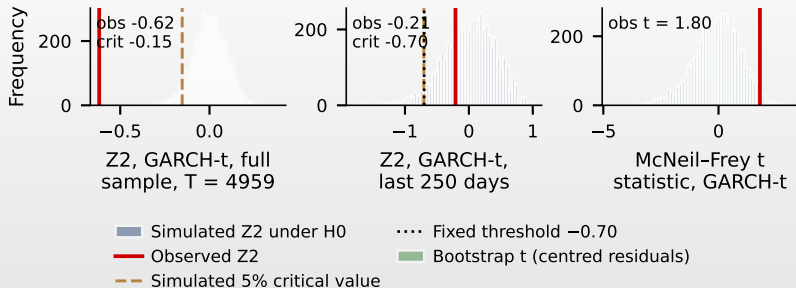
B3 [Solved]: Solution (1/2) — Step by Step and in Full

1. 12.03.2020: $e_t = (9.99 - 13.43)/4.81 = -0.71$ (the loss was below the forecast ES); contribution $0.744/123.98 = 0.0060$, out of 192 breach days.

Model	n	$\bar{e} / t$	p_{boot} / p_t	$Z_2 (T = 4959)$	$p / \text{crit.}$	$Z_2, p (250 \text{ d})$
GARCH-t	192	0.106 / 1.80	0.023 / 0.037	-0.618	<0.001 / -0.153	-0.210, 0.289
FHS	130	0.093 / 1.21	0.096 / 0.114	-0.091	0.154 / -0.150	0.142, 0.602
HS	156	0.363 / 2.88	<0.001 / 0.002	-0.405	<0.001 / -0.154	0.723, 0.975

- GARCH-t: $t = 1.80$, bootstrap $p = 0.023$ against Student- t $p = 0.037$: skewed residuals make the t approximation poor
- Full sample: GARCH-t ($Z_2 = -0.618$) and HS rejected, FHS passes; last 250 days: no model is rejected, GARCH-t has only 6 breaches
- FHS: $p = 0.154$, critical value -0.150 with 5000 draws (seed 11); the lecture uses 2000 draws (seed 1) and gets $p = 0.172$, critical value -0.152 : Monte Carlo variation, same decision

B3 [Solved]: Solution (2/2) — Null Distributions and Limits

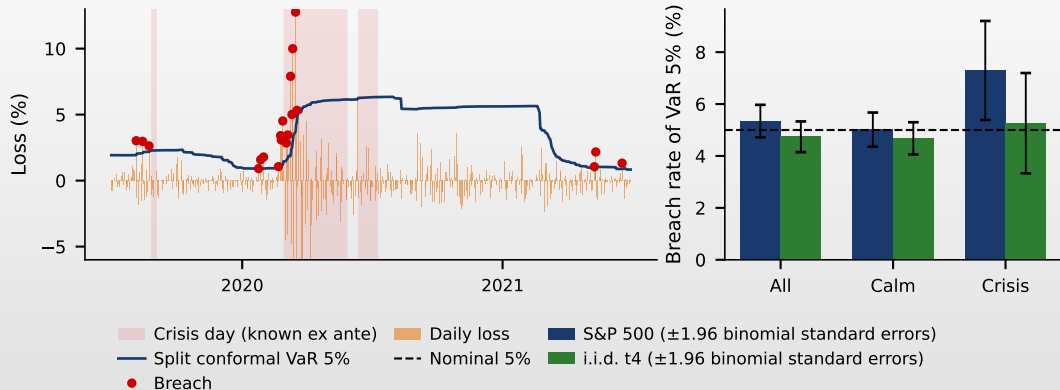


- The null distribution of Z_2 shrinks like $1/\sqrt{T\alpha}$: 5% critical value -0.153 on 4 959 days, -0.705 on 250 days (close to the reference -0.70)
- Interpretation: the GARCH-t rejection is evidence against the whole procedure (volatility filter, t tail, estimation); FHS, with the same filter, passes, which points to the tail but does not prove it; 5000 draws give a Monte Carlo standard error of about 0.3 points at $p = 5\%$

B7 [Solved]: Conformal VaR in Calm and Crisis Periods — Task

- **Question:** does split conformal VaR keep its coverage on crisis days of the S&P 500?
- **Inputs**
 - ▶ Timeline for forecast day t (all windows end at $t - 1$): scale $\hat{\sigma}_t$ = standard deviation of $r_{t-250}, \dots, r_{t-1}$; calibration scores $s_{t-250}, \dots, s_{t-1}$, $s_j = L_j / \hat{\sigma}_j$; crisis flag: standard deviation of $r_{t-20}, \dots, r_{t-1}$ above the 90th percentile of its values over the previous 1000 days
 - ▶ $\text{VaR}_t = \hat{\sigma}_t s_{(k)}$, $k = \lceil 251(1 - \alpha) \rceil$; evaluation 03.01.2007–18.09.2026 (4 959 days); benchmark: i.i.d. Student- t_4 returns $\times 1\%$, seed 5, same length
- **Tasks**
 1. Build VaR 5% and VaR 1% and the breach series on the S&P 500.
 2. Fill the grid (data source $\times$ level $\times$ all/calm/crisis) with rates and counts; add the Kupiec p (all days) and the Christoffersen independence p .
 3. Repeat on the i.i.d. benchmark series: its returns are exchangeable, but the scores standardised on a rolling window are only approximately exchangeable.
 4. Plot a VaR path with crisis shading and the rates with their binomial intervals.
- **Report:** the grid, two p -values per series, one figure
- **Interpretation**
 - ▶ is the marginal guarantee met on real data?
 - ▶ what changes between real and exchangeable data: coverage or independence?
- Notebook section: B7

B7 [Solved]: Solution (1/2)



B7 [Solved]: Solution (2/2)

- S&P 500, VaR 5%: all $265/4\,959 = 5.34\%$ (Kupiec $p = 0.272$), calm $213/4\,246 = 5.02\%$, crisis $52/713 = 7.29\%$; independence rejected ($p < 0.001$); VaR 1%: 1.29% overall ($p = 0.049$), 2.10% in crises
- i.i.d. benchmark: 4.74%, calm 4.68%, crisis 5.26% (27/513), independence not rejected ($p = 0.965$); the binomial intervals ignore dependence, so the real-data ones are too narrow
- Interpretation: the average is close to 5%, but volatility clustering breaks exchangeability, so breaches cluster and crisis coverage fails

 MFM_ch8_sem_conformal

B2 [Proposed]: Traffic Light for GARCH-t on the BET — Task

- **Question:** would the supervisor have penalised the GARCH-t model (VaR 1%) on the BET since 2007?
- **Worked example to follow:** A5, B1
- **Inputs**
 - ▶ GARCH-t VaR 1% and VaR 2.5% forecasts on the BET; evaluation 03.01.2007–18.09.2026 ($T = 4\,937$); the first complete 250-day window ends on 21.12.2007, so there are 4 688 windows; zones and multipliers: A5
- **Tasks**
 1. Count the VaR 1% exceptions in the last 250 days; give the zone, $P(X \leq x)$ and the FRTB multiplier.
 2. For every window, the count and zone; report the share of windows in each zone and the worst window with its end date.
 3. Count the VaR 2.5% exceptions in the last 250 days and compare with the desk limit of 30.
 4. Compute the full-sample breach rate and Kupiec p -value.
- **Interpretation:** is time in yellow evidence against the model? (overlapping windows: shares are descriptive, not a test)
- Notebook section: B2

Exit Questions and Team Project

Exit questions

- Which model would you provisionally choose for the S&P 500?
- On what evidence?
- What is the main limitation of that evidence?

Assessed team project (one)

- C1: a provisional VaR/ES recommendation for a Romanian trading book (extension pack 6), as a Quantlet folder in the team repository
- Rubric: data and timing correct (no look-ahead), minimum test set, evidence table, recommendation with its limitation
- Report actual AI use in AI_USE.md and every error you found in AI_ERRORS.md: request, answer, error, how found, correction; no minimum number of errors
- Extension packs are optional

Extension Packs: What Each One Needs

Pack	Exercises	Prerequisites
1. Test theory	A1 Extended [Solved], A2, A4	A1, A3
2. Scores and ES theory	A6 [Solved], A9, A10	A7, B5
3. Conformal VaR	A8 [Solved], B8	B7
4. More markets	B4, B6	B3, B5
5. Estimation risk	B9	B1; Chapter 8 , slide “Backtests Under Estimation Risk”
6. Projects	C1 (assessed), C3	B1–B5; C3 also needs R

- Each exercise has its own notebook section (C3 is solved in R; its notebook section gives only the setup); do them in order within a pack
- C1 and C3 are possible starts for the team project

A1 Extended [Solved]: Why LR_{uc} Is χ_1^2 — Task

- ▣ **Question:** where do the χ_1^2 limit and the noncentrality λ of A1 come from?
- ▣ **Inputs:** i.i.d. Bernoulli(π) hits, $\ell(\pi) = x \ln \pi + (T - x) \ln(1 - \pi)$, $\text{LR}_{uc} = 2[\ell(\hat{\pi}) - \ell(\alpha)]$; local alternatives $\pi_T = \alpha + h/\sqrt{T}$ with h fixed
- ▣ **Tasks**
 1. Expand $\ell(\alpha)$ to second order around $\hat{\pi}$ and show $\text{LR}_{uc} = T(\hat{\pi} - \alpha)^2/(\alpha(1 - \alpha)) + o_p(1)$ under H_0 .
 2. Use the CLT (central limit theorem: $\sqrt{T}(\hat{\pi} - \alpha)$ is approximately Normal) to conclude $\text{LR}_{uc} \xrightarrow{d} \chi_1^2$.
 3. Under π_T , derive the limit $\chi_1^2(\lambda)$ and give λ .
- ▣ **Report:** the three steps
- ▣ **Interpretation:** why is the local approximation poor at $\alpha T = 2.5$?
- ▣ Notebook section: A1

A1 Extended [Solved]: Solution

1. $\ell'(\hat{\pi}) = 0$ at the MLE; $\ell(\alpha) = \ell(\hat{\pi}) + \frac{1}{2}\ell''(\bar{\pi})(\alpha - \hat{\pi})^2$ with $\bar{\pi}$ between α and $\hat{\pi}$, and $-\ell''(\bar{\pi})/T = \frac{\hat{\pi}}{\bar{\pi}^2} + \frac{1-\hat{\pi}}{(1-\bar{\pi})^2} \xrightarrow{P} \frac{1}{\alpha(1-\alpha)}$.
 2. So $\text{LR}_{uc} = Z_T^2 + o_p(1)$, $Z_T = \sqrt{T}(\hat{\pi} - \alpha)/\sqrt{\alpha(1-\alpha)} \xrightarrow{d} N(0, 1)$ by the CLT: χ_1^2 , critical value 3.84.
 3. Under π_T : $\sqrt{T}(\hat{\pi} - \alpha) = \sqrt{T}(\hat{\pi} - \pi_T) + h$, so $Z_T \xrightarrow{d} N(h/\sqrt{\alpha(1-\alpha)}, 1)$ and $\text{LR}_{uc} \xrightarrow{d} \chi_1^2(\lambda)$, $\lambda = h^2/(\alpha(1-\alpha)) = T(\pi - \alpha)^2/(\alpha(1-\alpha))$.
- Interpretation: the expansion needs many breaches; with 2.5 expected, x takes few integer values, so its distribution is strongly discrete, hence the gap between 36% (local) and 24% (exact) in A1

A2 [Proposed]: How Many Days to Detect a Doubled Rate? — Task

- ▣ **Question:** how many days are needed to detect, with 80% power, a VaR 2.5% whose true breach rate is 5%?
- ▣ **Worked example to follow:** A1
- ▣ **Inputs:** $\alpha = 0.025$, $\pi = 0.05$, the rule “reject if $LR_{uc} > 3.84$ ”; the FRTB desk limit of 30 exceptions of VaR 2.5% is a separate regulatory rule
- ▣ **Tasks**
 1. At $T = 250$: λ , the local power, the exact acceptance region, the exact power and the actual size.
 2. Find λ^* with $P(\chi_1^2(\lambda^*) > 3.84) = 0.80$ by root finding and solve for the asymptotic T .
 3. For $T = 100, \dots, 700$: acceptance set $\{x : LR_{uc}(x) \leq 3.84\}$, power = $1 - P(X \in \text{set} \mid \pi)$; plot power against T and find the first T with power $\geq 80\%$.
 4. Check whether the power stays above 80% for every larger T .
 5. Compute $P(X > 30)$ for $X \sim \text{Binomial}(250, 0.05)$.
- ▣ **Report:** λ , powers, size, λ^* , the two values of T , $P(X > 30)$, one figure
- ▣ **Interpretation:** what does the desk limit screen for?
- ▣ Notebook section: A2

A4 [Proposed]: LR versus DQ at $x = 7$ — Task

- ▣ **Question:** why do LR_{uc} and the intercept-only DQ statistic give different numbers for the same breaches?
- ▣ **Worked example to follow:** A3
- ▣ **Inputs**
 - ▶ Hit: $T \times 1$ vector of $I_t - \alpha$; X : $T \times k$ regressors known at $t - 1$, full rank; $\text{Hit} = X\beta + u$; $Q = \text{plim } X'X/T$ (plim: probability limit as $T \rightarrow \infty$) positive definite ([Engle & Manganelli \(2004\)](#))
 - ▶ Example design with one lag: rows $X_t = (1, \text{Hit}_{t-1})$, so $k = 2$
- ▣ **Tasks**
 1. Show $E[\text{Hit}_t \mid \mathcal{F}_{t-1}] = 0$ and $\text{Var}(\text{Hit}_t \mid \mathcal{F}_{t-1}) = \alpha(1 - \alpha)$ under H_0 .
 2. With $X_t = 1$ show $DQ = T(\hat{\pi} - \alpha)^2 / (\alpha(1 - \alpha))$; compute it for $T = 250$, $x = 7$ and compare with $LR_{uc} = 5.50$ ($p = 0.019$).
 3. Compute both statistics for $x = 0, \dots, 15$ and their acceptance sets at 3.84.
 4. Advanced: derive $DQ \xrightarrow{d} \chi_k^2$ from a martingale-difference CLT for $T^{-1/2}X'\text{Hit}$.
- ▣ **Report:** the two statistics and p -values, the acceptance sets, the derivation
- ▣ **Interpretation:** do the two tests disagree in their decision or only in their numbers?
- ▣ Notebook section: A4

A6 [Solved]: Consistency of the Pinball and FZ0 Losses — Task

▣ Questions

- ▶ why does the truth minimise the expected pinball and FZ0 scores?
- ▶ does a simulation confirm it?

▣ Inputs

- ▶ L with $E|L| < \infty$, density $f > 0$ near VaR_α , $\text{ES}_\alpha > 0$; forecasts $e \geq v > 0$; $\alpha = 2.5\%$
- ▶ Simulation: $L = \sqrt{(4-2)/4} Z$, $Z \sim t_4$, 10^6 draws, seed 5; truth $\text{VaR} = 1.963$, $\text{ES} = 2.824$ (unit-variance t_4 formulas)

▣ Tasks

1. Differentiate $E[(1\{L > v\} - \alpha)(L - v)]$ in v , find the zero and show it is a minimum.
2. For $\text{FZ0}(v, e; L) = \frac{1\{L > v\}(L - v)}{\alpha e} + \frac{v}{e} + \ln e - 1$ compute ∂_v and ∂_e of the expectation and show that the true pair is the unique minimiser.
3. Minimise the average pinball and FZ0 losses on the simulated draws and plot the score curve and the FZ0 contours.

▣ **Report:** derivatives, the argument for the minimum, numerical minimisers, one figure

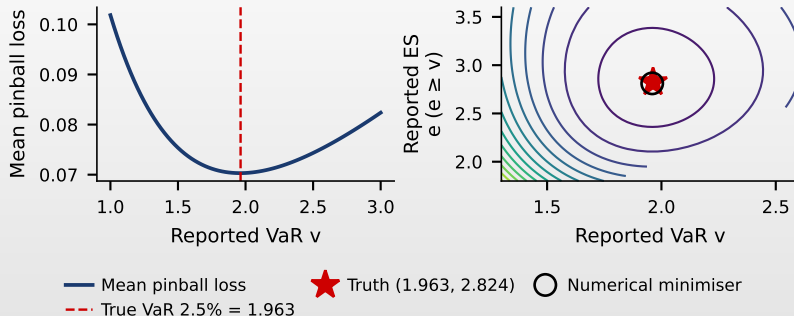
▣ **Interpretation:** why does consistency make ranking by average score meaningful?

▣ Notebook section: A6

A6 [Solved]: Solution (1/2) — Derivations

1. $\frac{d}{dv} E[(1\{L > v\} - \alpha)(L - v)] = F(v) - (1 - \alpha)$: negative below and positive above VaR_α , zero at $F(v) = 1 - \alpha$; second derivative $f(v) > 0$: a unique minimum.
2. $\partial_v E[\text{FZ0}] = \frac{1}{e} (1 - \frac{P(L > v)}{\alpha})$: same sign pattern, zero only at $v = \text{VaR}_\alpha$, for every e .
3. At that v , $\partial_e E[\text{FZ0}] = \frac{1}{e^2} (e - v - E[(L - v)1\{L > v\}]/\alpha) = \frac{1}{e^2} (e - \text{ES}_\alpha)$: negative for $e < \text{ES}_\alpha$, positive above: the profiled objective in e has its unique minimum at ES_α .
4. Identification functions: $V_1 = \alpha - 1\{L > v\}$, $V_2 = e - v - \frac{1}{\alpha} 1\{L > v\}(L - v)$; their means vanish only at the true pair (Fissler & Ziegel (2016)).

A6 [Solved]: Solution (2/2) — Numerical Check



- Minimisers: pinball $v = 1.960$; FZ0 $(v, e) = (1.960, 2.812)$; truth $(1.963, 2.824)$: equal up to simulation error
- Interpretation: a forecaster who knows F minimises the expected score by reporting the truth, so a lower average score is evidence of a better forecast

A9 [Proposed]: ES Alone Is Not Elicitable — Task

- ▣ **Question:** why can no score of ES alone reward honest forecasts?
- ▣ **Worked example to follow:** A6
- ▣ **Inputs**
 - ▶ $T(F)$ is elicitable if $T(F) = \arg \min_x E_F[S(x, L)]$ with a unique minimiser ([Gneiting \(2011\)](#)); δ_c : the point mass at c (loss c for sure)
 - ▶ ES for any distribution, atoms included: $ES_\alpha = \frac{1}{\alpha} \int_{1-\alpha}^1 q_u du$, the mean of the upper- α tail quantiles of the loss (an atom may be split)
- ▣ **Tasks**
 1. Show that if T is elicitable, every level set $\{F : T(F) = t\}$ is closed under mixtures $(1 - \lambda)F_0 + \lambda F_1$.
 2. $F_0 = \delta_1$; F_1 : $P(L = 2) = \alpha/2$, $P(L = 0) = 1 - \alpha/2$: compute ES_α of F_0 , F_1 and $\frac{1}{2}F_0 + \frac{1}{2}F_1$ from the tail-quantile picture.
 3. Explain, conceptually, why this does not contradict the joint elicibility of (VaR, ES).
- ▣ **Report:** the argument, three ES values, one sentence
- ▣ **Interpretation:** what does this imply for ranking ES forecasts?
- ▣ Notebook section: A9

A10 [Proposed]: Moments of the Cumulative Violation — Task

▣ Questions

- ▶ what are the null moments of H_t ?
- ▶ is the Normal critical value adequate at $T = 250$?

▣ Worked example to follow: A7

- ▣ **Inputs:** PITs u_t i.i.d. $U(0, 1)$ under a correct forecast; $H_t = \frac{1}{\alpha}(u_t - (1 - \alpha))1\{u_t > 1 - \alpha\}$ (Du & Escanciano (2017)): zero on 97.5% of days, rising linearly to 1 in the tail; $\alpha = 2.5\%$

▣ Tasks

1. Plot $H(u)$ for $u \in (0.9, 1)$.
2. Compute $E[H]$ and $E[H^2]$ by integrating over $(1 - \alpha, 1)$; deduce $\text{Var}(H) = \alpha(1/3 - \alpha/4)$.
3. Check both moments on $4 \cdot 10^6$ uniform draws (seed 6).
4. Compute the one-sided Normal 5% cut-off for $\bar{H}$ at $T = 250$; simulate 20 000 samples of 250 PITs (seed 12) and compare with their 95th percentile.

- ▣ **Report:** the moments, the check, both cut-offs and the actual size of the Normal cut-off

- ▣ **Interpretation:** why does H_t carry more information than I_t ?

- ▣ Notebook section: A10

A8 [Solved]: Conformal Quantile and One ACI Step — Task

- Question: how does split conformal VaR 10% react to one breach under ACI?

- Inputs

- ▶ $n = 19$ calibration scores (sorted, table); tomorrow's $\hat{\sigma} = 1.4\%$; target $\alpha = 0.10$ (fixed); ACI step $\gamma = 0.02$, $\alpha_1 = 0.10$ (Gibbs & Candès (2021))
- ▶ For this one-step illustration the scores and $\hat{\sigma}$ are frozen; if $k > n$ the quantile is $+\infty$

- Tasks

1. Compute $k = \lceil (n+1)(1-\alpha) \rceil$ ($\lceil \cdot \rceil$: round up), $s_{(k)}$ (the k -th smallest score) and $\text{VaR} = \hat{\sigma} s_{(k)}$.
2. Compute $\alpha_2 = \alpha_1 + \gamma(\alpha - \text{err}_1)$ after a breach and after no breach.
3. Recompute k and VaR for both values of α_2 ; below which α_t does the VaR become infinite?

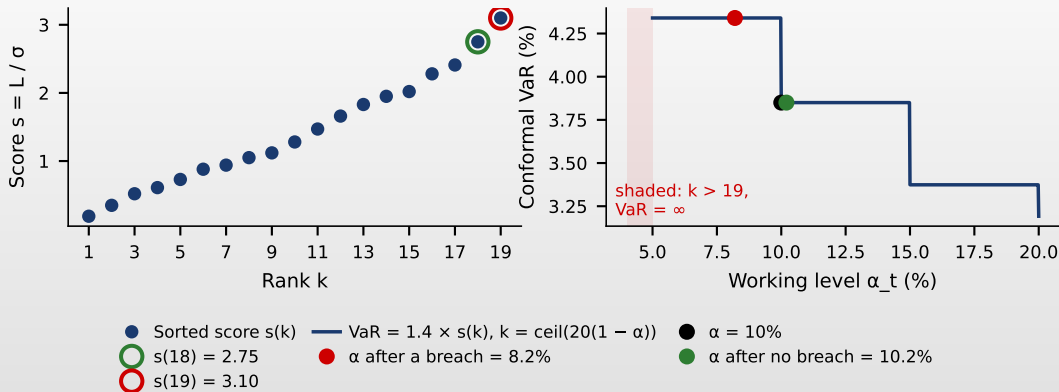
- Report: k , three VaR values, the infinite-VaR threshold

- Interpretation: how smoothly does ACI move the VaR?

- Notebook section: A8

k	$s_{(k)}$	k	$s_{(k)}$	k	$s_{(k)}$
1	0.19	8	1.05	15	2.02
2	0.35	9	1.12	16	2.28
3	0.52	10	1.28	17	2.41
4	0.61	11	1.47	18	2.75
5	0.73	12	1.66	19	3.10
6	0.88	13	1.83		
7	0.94	14	1.95		

A8 [Solved]: Solution (1/2)



A8 [Solved]: Solution (2/2)

- $k = \lceil 20 \cdot 0.9 \rceil = 18$; $s_{(18)} = 2.75$; $\text{VaR} = 1.4 \cdot 2.75 = 3.85\%$
- After a breach: $\alpha_2 = 0.10 + 0.02(0.10 - 1) = 0.082$, $k = \lceil 18.36 \rceil = 19$, $\text{VaR} = 1.4 \cdot 3.10 = 4.34\%$
- No breach: $\alpha_2 = 0.102$, $k = \lceil 17.96 \rceil = 18$: the VaR does not move; for $\alpha_t < 1/(n+1) = 5\%$, $k = 20 > n$: $\text{VaR} = \infty$
- Interpretation: with small n ACI moves in jumps of one order statistic; small changes of α_t are invisible until they cross a grid point $j/(n+1)$

 MFM_ch8_sem_conformal

B8 [Proposed]: Conformal VaR on the BET and Bitcoin — Task

- **Question:** does ACI repair crisis coverage on the BET and on Bitcoin, or only the long-run average?
- **Worked example to follow:** B7
- **Inputs**
 - ▶ BET daily closes from 03.01.2007 and Bitcoin daily closes (24:00 UTC) from 14.06.2017, to 18.09.2026; VaR 5%: split conformal, ACI and GARCH-t; scores, windows and crisis flag as in B7
 - ▶ ACI: $\alpha_1 = 0.05$, $\gamma = 0.005$, started on the first evaluation day; an infinite VaR counts as no breach and is reported separately
- **Tasks**
 1. For each market and method, report breach counts and rates on all, calm and crisis days.
 2. Report the number of days with an infinite ACI VaR.
 3. Attach a binomial 95% interval to each crisis rate and say which differences from 5% are clear.
- **Report:** a rate-and-count table, the infinite-VaR count, one figure
- **Interpretation:** an evidence-qualified comparison of the three methods, not a single winner
- Notebook section: B8

B4 [Proposed]: Z_2 by Simulation on the BET and Bitcoin — Task

- ▣ **Question:** are the ES 2.5% forecasts of GARCH-t and GARCH-EVT calibrated on the BET and on Bitcoin?
- ▣ **Worked example to follow:** B3
- ▣ **Inputs**
 - ▶ Forecast panels of the BET (from 2007) and Bitcoin (from 2017); Algorithm 2 of B3 with 5000 draws, seed 11, forecasts held fixed
 - ▶ Samplers: GARCH-t as in B3; GARCH-EVT $L^* = -\mu_t + \sigma_t \left(u + \frac{\beta}{\xi} \left[\left(\frac{1-u}{p_u} \right)^{-\xi} - 1 \right] \right)$ with the stored GPD threshold u , shape ξ , scale β and share p_u of standardised losses above u
- ▣ **Tasks**
 1. On the full sample, build a table with one row per market and model (BET and Bitcoin $\times$ GARCH-t and GARCH-EVT): Z_2 , p , 5% critical value, breach days.
 2. Repeat on the last 250 days.
 3. Plot the four null distributions with the observed Z_2 .
- ▣ **Report:** the eight-row table and one figure
- ▣ **Interpretation:** a provisional comparison with its limitations: why does the verdict change between the full sample and the last year?
- ▣ Notebook section: B4

B6 [Proposed]: Comparing Models on Bitcoin and EUR/RON — Task

- ▣ **Question:** can the data separate the six models on Bitcoin and on EUR/RON?
- ▣ **Worked example to follow:** B5
- ▣ **Inputs**
 - ▶ Daily FZ0 scores of the six models on Bitcoin and on EUR/RON; DM with Newey–West lags $\lfloor 4(T/100)^{2/9} \rfloor$
 - ▶ MCS (Hansen, Lunde & Nason (2011)): T_{max} statistic (the largest standardised mean loss differential among the models still in the set), circular block bootstrap, blocks of 10 days, 1000 replications, seed 2, 90% set
 - ▶ Murphy diagram (Ehm, Gneiting, Jordan & Krüger (2016)): Bitcoin VaR 2.5%, HS versus GARCH-EVT, 240 thresholds from 2.8% to 17.2%, pointwise HAC bands
- ▣ **Tasks**
 1. Find the lowest mean FZ0 in each market and run exploratory DM tests of every other model against it.
 2. Run the MCS; show the first elimination step (statistic, p -value, model removed) and list the 90% set.
 3. Plot the Murphy difference with its bands and say whether the curves cross.
- ▣ **Report:** a DM/MCS table, the elimination steps, one figure
- ▣ **Interpretation**
 - ▶ which models remain in the MCS for each market?
 - ▶ what limits the separation?
- ▣ **Notebook section:** B6

B9 [Proposed]: Kupiec Size Under Estimation Risk — Task

- ▣ **Question:** how often does Kupiec reject a correct GARCH-t model when its parameters are estimated?
- ▣ **Worked example to follow:** B1
- ▣ **Inputs**
 - ▶ Fixed DGP: GARCH(1,1)-t fitted to the S&P 500 1990–2026 ($a = 0.102$, $b = 0.892$, $\nu = 6.1$); 400 paths of $R + 5000$ days after a burn-in of 500, seed 9
 - ▶ Fixed scheme: estimate once on the first $R = 1000$ days (ν floored at 2.2), then forecast VaR 1% for $P = 250, 1000, 5000$ days; on the same path also the true-parameter VaR
- ▣ **Tasks**
 1. Record the Kupiec rejection at 5% for the estimated and the true forecasts, paired on each path.
 2. Report each rejection rate with its own Monte Carlo standard error $\sqrt{\hat{p}(1 - \hat{p})/400}$, $\hat{p}$: the rejection rate.
 3. Compare the standard deviation of the breach rate across paths with $\sqrt{\alpha(1 - \alpha)/P}$; plot the rates against P/R .
 4. Going further: an actual bootstrap calibration: simulate under the fitted null, re-estimate on each replicate and take the 95th percentile of LR_{uc} as the critical value.
- ▣ **Report:** a table (P , two rates with standard errors, two standard deviations) and one figure
- ▣ **Interpretation:** how does the distortion grow with P/R (Escanciano & Olmo (2010))?
- ▣ Notebook section: B9

C1 [Proposed]: A Provisional Model for a Romanian Trading Book — Task (1/2)

- **Question:** which VaR/ES model would you provisionally recommend for a portfolio in lei, half in the BET and half in euro?
- **Inputs**
 - ▶ BET daily closes and the BNR EUR/RON reference rate; align the **prices** on common trading days, then simple returns, then the portfolio log return $r_t^p = \ln(1 + \frac{1}{2}R_t^{BET} + \frac{1}{2}R_t^{EUR})$ (R : simple returns), daily rebalanced, no costs
 - ▶ The six models of [Chapter 8](#), 1000-day window; evaluation from 09.07.2009 (after 2008: the 2008 crisis is an unobserved stress scenario) to 18.09.2026
- **Tasks**
 1. Build the portfolio and plot its losses with the VaR 1% of HS, FHS and GARCH-EVT.
 2. Run the minimum test set (next slide) for all six models.
- Notebook section: C1

C1 [Proposed]: Evidence and Recommendation — Task (2/2)

▣ Questions

- ▶ what does the evidence support?
- ▶ what does it not support?

▣ Worked example to follow: B1–B5

▣ Inputs: minimum test set: frequency (Kupiec, CC), timing (durations or DQ), ES (Z_2 by simulation), comparison (FZ0, DM, 90% MCS), regulation (worst and last 250-day windows, desk limits)

▣ Tasks

1. Fill the evidence table: model $\times$ rate, p_{cc} , p_{dur} , Z_2 / p , worst window, last window (VaR 1% / VaR 2.5%), FZ0 / MCS p .
2. State the size and power of this set of tests at $T = 250$ (B1, A1).
3. Write one provisional recommendation with a challenger model, and list what FRTB approval would also require (desk P&L backtests, P&L attribution).

▣ Report: a Quantlet folder and one slide: figure, evidence table, recommendation, main limitation

▣ Interpretation: would the choice survive a year like 2008 or 2020?

▣ Notebook section: C1

C3 [Proposed]: Backtesting ES with ES Forecasts Only — Task (1/2)

- **Question:** do the regression-based ES backtests of [Bayer & Dimitriadis \(2022\)](#) agree with Z_2 and McNeil–Frey on the [Chapter 8](#) forecasts?
- **Inputs**
 - ▶ The ES 2.5% forecasts of the six models in the four markets (24 series); returns convention of the paper: $r_t = -L_t$, $\hat{e}_t = -\widehat{ES}_t < 0$
 - ▶ R (4.x) with the authors' package: `esback::esr_backtest(r, e, alpha = 0.025, version = v)`; $v = 1$: Strict ESR (two-sided), $v = 3$: Intercept ESR (one-sided $H_0: \gamma_1 \geq 0$, γ_1 : intercept of $r_t - \hat{e}_t$, negative when ES is underestimated); misspecification-robust covariance (default)
- **Tasks**
 1. Work one case first: S&P 500, GARCH-t: print the coefficients, their standard errors, the statistic and the p -value of both versions.
 2. Run both versions on all 24 series and draw a market-by-model heat map of the p -values.
 3. Compare the decisions at 5% with Z_2 and McNeil–Frey ([Chapter 8](#), section “Backtesting Expected Shortfall”).
- Notebook section: C3

C3 [Proposed]: Blocks and Size — Task (2/2)

▣ Questions

- ▶ how do the tests behave on one-year blocks?
- ▶ what is their size at $T = 250$?

▣ Worked example to follow: B3

- ▣ **Inputs:** non-overlapping 250-day blocks of the S&P 500 from the first evaluation day (drop the incomplete last block); the EGARCH(1,1)-t process of Eq. 2.1 of the paper, 1000 paths, a fixed seed

▣ Tasks

1. Report the share of blocks in which each model is rejected, and the number of failed fits (excluded from the share).
2. Simulate 1000 paths of 250 days, backtest the true ES 2.5% with the Intercept ESR, two-sided and one-sided, and report the rejection rates with their Monte Carlo standard errors.
3. Compare the two-sided rate with the 0.04 of Table 1 (robust covariance) and the one-sided rate with the nominal 5% (Table 2 reports it only at $T = 2500$).

- ▣ **Report:** a Quantlet folder and two slides: the decision heat map with Z_2 and McNeil–Frey, and the block shares with the size check

▣ Interpretation

- ▶ when ESR and Z_2 disagree, which would you trust as a supervisor who receives only ES forecasts?
- ▶ why would you trust it?

References (1/4)

- ▣ Acerbi, C., Szekely, B. (2014). [Backtesting expected shortfall](#). MSCI Inc., December 2014.
- ▣ Basel Committee on Banking Supervision (1996). [Supervisory framework for the use of “backtesting” in conjunction with the internal models approach to market risk capital requirements](#). Bank for International Settlements.
- ▣ Basel Committee on Banking Supervision (2019). [Minimum capital requirements for market risk \(rev. February 2019\)](#). Bank for International Settlements.
- ▣ Bayer, S., Dimitriadis, T. (2022). [Regression-based expected shortfall backtesting](#). *Journal of Financial Econometrics*, 20(3), 437–471.
- ▣ Christoffersen, P.F. (1998). [Evaluating interval forecasts](#). *International Economic Review*, 39(4), 841–862.
- ▣ Christoffersen, P., Pelletier, D. (2004). [Backtesting value-at-risk: a duration-based approach](#). *Journal of Financial Econometrics*, 2(1), 84–108.

References (2/4)

- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Du, Z., Escanciano, J.C. (2017). Backtesting expected shortfall: accounting for tail risk. *Management Science*, 63(4), 940–958.
- ▣ Ehm, W., Gneiting, T., Jordan, A., Krüger, F. (2016). Of quantiles and expectiles: consistent scoring functions, Choquet representations and forecast rankings. *Journal of the Royal Statistical Society, Series B*, 78(3), 505–562.
- ▣ Engle, R.F., Manganelli, S. (2004). CAViaR: conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4), 367–381.
- ▣ Escanciano, J.C., Olmo, J. (2010). Backtesting parametric value-at-risk with estimation risk. *Journal of Business & Economic Statistics*, 28(1), 36–51.
- ▣ Fissler, T., Ziegel, J.F. (2016). Higher order elicibility and Osband's principle. *Annals of Statistics*, 44(4).

References (3/4)

- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Giacomini, R., White, H. (2006). *Tests of conditional predictive ability*. *Econometrica*, 74(6), 1545–1578.
- ▣ Gibbs, I., Candès, E. (2021). *Adaptive conformal inference under distribution shift*. *Advances in Neural Information Processing Systems*, 34; arXiv:2106.00170.
- ▣ Gneiting, T. (2011). *Making and evaluating point forecasts*. *Journal of the American Statistical Association*, 106(494), 746–762.
- ▣ Hansen, P.R., Lunde, A., Nason, J.M. (2011). *The model confidence set*. *Econometrica*, 79(2), 453–497.
- ▣ Kupiec, P.H. (1995). *Techniques for verifying the accuracy of risk measurement models*. *Journal of Derivatives*, 3(2), 73–84.

References (4/4)

- ▣ McNeil, A.J., Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach. *Journal of Empirical Finance*, 7(3–4), 271–300.
- ▣ Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- ▣ Patton, A.J., Ziegel, J.F., Chen, R. (2019). Dynamic semiparametric models for expected shortfall (and value-at-risk). *Journal of Econometrics*, 211(2), 388–413.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). Finite-sample precision limits for expected shortfall forecast comparisons. *Mathematics*, 14(13), 2316.